Goodness of Estimators

Learning Outcomes

- Consistency
- Sufficiency
- Information
- Efficiency

Consistency

An estimator is considered a consistent estimator of θ if the estimator, on average, converges to θ as $n \to \infty$.

Consistency

Let $X_1, \ldots, X_n$ be a random sample from a distribution with parameter θ. The estimator $\widehat{\theta}$ is a consistent estimator of the θ if

1. $E\{(\widehat{\theta} - \theta)^2\} \to 0$ as $n \to \infty$
2. $P(|\widehat{\theta} - \theta| \geq \epsilon) \to 0$ as $n \to \infty$ for every $\epsilon > 0$

Sufficiency

Sufficiency evaluates whether a statistic (or estimator) contains enough information of a parameter θ. In essence a statistic is considered sufficient to infer θ if it provides enough information about θ.

Sufficiency

Let $X_1, \dots, X_n$ be a random sample from a distribution with parameter θ. A statistic $T = t(X_1, \dots, X_n)$ is said to be sufficient for making inferences of a parameter θ if condition joint distribution of $X_1, \dots, X_n$ given $T = t$ does not depend on θ.

Factorization Theorem

The **Factorization Theorem** provides a condition for a statistic $T(X)$ to be sufficient for a parameter θ given a probability density function or probability mass function.

Factorization Theorem

Let $X = (X_1, X_2, \dots, X_n)$ be a random sample with joint probability density (or mass) function $f(x|\theta)$, where θ is a parameter.

Factorization Theorem

Theorem: A statistic $T(X)$ is sufficient for θ if and only if the joint density (or mass) function $f(x|\theta)$ can be factored into the form

$$f(x|\theta) = g(T(x), \theta) \cdot h(x)$$

where:

- $g(T(x), \theta)$ is a function that depends on $T(x)$ and θ,
- $h(x)$ is a function that does not depend on θ.

Factorization Theorem

In other words, $f(x|\theta)$ can be written as a product of two functions, where only one function depends on the parameter θ and the sufficient statistic $T(X)$.

Implications: The Factorization Theorem is useful for identifying sufficient statistics, which summarize all necessary information from a sample about the parameter θ.

Information

In Statistics, information is thought of as how much does the data tell you about a parameter θ. In general, the more data is provided, the more information is provided to estimate θ.

Information

Information can be quantified using Fisher's Information $I(\theta)$. For a single observation, Fisher's Information is defined as

$$I(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\log\{f(X;\theta)\}\right],$$

$\sqrt{Var(\hat{\theta})}$ Used for MLE's

where $f(X;\theta)$ is either the PMF or PDF of the random variable X.

Information

Furthermore, $I(\theta)$ can be defined as

$$I(\theta) = Var\left\{\frac{\partial}{\partial\theta}\log f(X;\theta)\right\}.$$

Proof

Show the following property:

$$E\left[-\frac{\partial^2}{\partial\theta^2}\log\{f(X;\theta)\}\right] = Var\left\{\frac{\partial}{\partial\theta}\log f(X;\theta)\right\}$$

$$\frac{d}{d\theta} E\left(\frac{d}{d\theta} \log f(x;\theta)\right) = 0$$

$$\frac{d}{d\theta} \int \frac{d}{d\theta} \log f(x;\theta)\, f(x;\theta)\, dx$$

$$\int \frac{d}{d\theta} \left[\frac{d}{d\theta} \log f(x;\theta)\, f(x;\theta)\right] dx$$

$$\int \frac{d^2}{d\theta^2} \log f(x;\theta)\, f(x;\theta)$$

$$+ \frac{d}{d\theta} \log f(x;\theta)\, \frac{d}{d\theta} f(x;\theta)\, dy$$

$$\int \frac{d^2}{d\theta^2} \log f(x;\theta)\, f(x;\theta)\, dx$$

$$+ \int \frac{d}{d\theta} \log f(x;\theta)\, \frac{d}{d\theta} f(x;\theta)\, dx$$

$$\int \frac{d^2}{d\theta^2} \log f(x;\theta)\, f(x;\theta)\, dx$$

$$E\left(\frac{d^2}{d\theta^2} \log f(x;\theta)\right) +$$

$$\int \underline{\frac{d}{d\theta} \log f(x;\theta) \frac{d}{d\theta} f(x;\theta)}\, dx$$

$$\int \frac{d}{d\theta} \log f(x;\theta) \frac{\frac{d}{d\theta} f(x;\theta)}{f(x;\theta)} f(x;\theta)\, dx$$

$$\int \frac{d}{d\theta} \log f(x;\theta) \; \frac{d}{d\theta} \log f(x;\theta) \, f(x;\theta)\, dx$$

$$\int \left[\frac{d}{d\theta} \log f(x;\theta)\right]^2 f(x;\theta)\, dx$$

$$E\left(\frac{d^2}{d\theta^2} \log f(x;\theta)\right) + E\left(\left[\frac{d}{d\theta} \log f(x;\theta)\right]^2\right)$$

$$E(x^2) = Var(x) + E(x)^2$$

$$E\left[\frac{d}{d\theta} \log f(x;\theta)\right]$$

$$\int \frac{d}{d\theta} \log f(x;\theta)\, f(x;\theta)\, dx$$

$$\int \frac{d}{d\theta} f(x;\theta) \cancel{\frac{f(x;\theta)}{f(x;\theta)}}\, dx$$

$$\int \frac{d}{d\theta} f(x;\theta)\, dx$$

$$\frac{d}{d\theta}\int f(x;\theta)\,dx$$

$$\frac{d}{d\theta}[1] = 0$$

$$E\left(\frac{d}{d\theta}\log f(x;\theta)\right] = 0$$

$$E\left(\left[\frac{d}{d\theta}\log f(x;\theta)\right]^2\right) = Var\left(\frac{d}{d\theta}\log f(x;\theta)\right)$$

$$E\left(\frac{d^2}{d\theta^2}\log f(x;\theta)\right) + Var\left(\frac{d}{d\theta}\log f(x;\theta)\right) = 0$$

$$E\left(\frac{d^2}{d\theta^2}\log f(x;\theta)\right) = -Var\left(\frac{d}{d\theta}\log f(x;\theta)\right)$$

$$E\left(-\frac{d^2}{d\theta^2}\log f(x;\theta)\right) = Var\left(\frac{d}{d\theta}\log f(x;\theta)\right)$$

$$\frac{d}{d\theta}E\left(\frac{d}{d\theta}\log f(x;\theta)\right) = 0$$

Efficiency

The efficiency of an estimator T, where T is an unbiased estimator of θ, is defined as

$$efficiency\, of\, T = \frac{1}{Var(T)nI(\theta)}$$